Independent ordinary Kummer fiber and orientation review

Task T-1431. 3 November 2026.

Ordinary fiber	Opposite coefficient pair containment
Projection / normalization	Quintic identity; explicit solve premises
Neighbour orientation	Abstract group / fiber / anchor premises
Fresh machine review	12 producer + 5 reviewer declarations

Exact scoped proofs reviewed; native solve4 and geometric/label bridges remain separate. T-1431.

Fresh Lean verification passed all twelve named ordinary-fiber, extraction, chart and orientation declarations, plus five reviewer-authored boundary controls. Exact named transitive axiom readbacks use only propext, Classical.choice and Quot.sound. A deliberately false theorem exited1 and three native checker rejection controls passed. Independent source fidelity is accepted for the stated conditional claims.

Coefficient fiber. The two valid equations are exactly the constant and linear remainders of (b0+b1 X)^2 modulo X^2+a1 X+a0. Shared squared slope and those equations give two opposite coefficient pairs at most, including the zero-slope case. This is an algebraic coefficient predicate; it does not parse arbitrary native divisors or prove their geometric validity.

Projection and chart. The source formulas match pruned.rs: q=(1,-a1,a0,xi4), and C1 row i is scale[i] times (m0[i],m1[i],m2[i],-1). The general inverse routine's apparent transpose constructs exactly these rows. The extracted square equals b1 squared plus f6 times (a1 squared minus a0) squared; only the quintic f6=0 specialization removes that term. Source curve construction begins with X and multiplies four linear factors, consistent with degree5. The formal normalization conclusion assumes an injective C1 map, a correct solve equation and nonzero projective scale. Native solve4 pivot search, swaps, normalization and elimination are not proved by this packet.

Orientation. The group theorem correctly cancels the two sign alternatives when twice the anchor is nonzero and the neighbour projection has the stated plus/minus fiber. The coefficient wrapper assumes a representation respecting negation. Its conclusion is equality of represented group elements; it does not prove a concrete Jacobian representation, actual Kummer neighbour fiber, geometric transition action, prime-order anchor or walk-label invariant. Those remain explicit source bridges.

Boundary controls. Fresh reviewer declarations establish a sextic residual counterexample, failure of zero-scale normalization, sign ambiguity with a two-torsion anchor, loss of orientation with a constant projection lacking the fiber premise, and a valid zero-slope ordinary pair. These are exact countermodels and a positive example, not sampled curve coverage. The first replay stopped before theorem checking because a thread could not be created under a4GiB address-space cap. Its error is retained. The corrected bounded replay used8GiB address space,3.5GiB Lean heap and16MiB thread stack, with unchanged proof sources.

Audit reconciliation. The older goal-coverage-audit correctly identified independent review of these twelve declarations as pending; this packet supplies that review. The later native-branch-composition packet already independently reviewed the literal word row fold and nonzero preservation under sixteen matrix-square certificates. It does not establish solve4 correctness. No later general solve4 proof was found in the inspected packet inventory. Next construct an augmented4x5 elimination invariant, with source pivot/swap/eliminate order, canonical-word assumptions and returned equation, then connect successful C1 solve and chart guards to this normalization theorem. Preserve the105 saved matrix records and N19 cohort; neither was rerun.

Independence and trust. This reviewer authored none of the producer theorem/native source. Producer source/report were read to select scope, so this is non-author source review, not blind derivation. Local imported source and compiled objects were copied and hash-bound before/after verification; upstream Lean4.34.1/Mathlib compiled cache trust remains, and no independent binary kernel rechecker ran. Review changes no official hypothesis or scientific admission. No scientific trajectory, cloud work or new timing comparison occurred.